

Degree: B.Tech., Stream: ALL (Group- A)
Term - I, First Semester Examination, August - 2025
Subject Code: ESCEE101
Subject Name: BASIC ELECTRICAL ENGINEERING

Full Marks: 30

Duration: 1 Hour

Part - A

Attempt any 5 questions

Each question carries 2 marks (2×5)

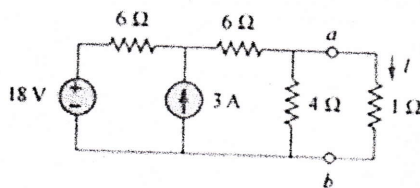
1. State and explain Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL) with the help of a simple circuit diagram. 2
2. Critically compare the advantages and limitations of Thevenin's and Norton's theorems in practical circuit analysis. 2
3. Can DC be applied to a transformer? Justify your answer. 2
4. What is the purpose of laminating the core in a transformer? 2
5. What are the key differences between core-type and shell-type transformers? 2
6. A 12 V battery supplies 3 A current. Find the total power delivered. 2

Part - B

Attempt any 2 questions

Each question carries 5 marks (5×2)

7. Using Thevenin's theorem, find the Thevenin's equivalent circuit between the terminal a and b and find I. 5



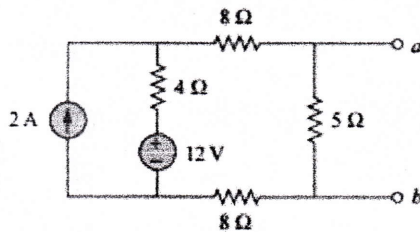
8. A 400/2500 V, 50 kVA single phase transformer working in an industrial application, determine the primary and secondary current of the transformer. Also determine the number of turns of the secondary side if the primary number of turns is 250. 5
9. State and explain Thevenin's theorem its different steps for solving a circuit using. What is the formula to calculate load current. 5

Part - C

Attempt any 1 question

Each question carries 10 marks (10 × 1)

10. Find the Norton equivalent circuit of the circuit below at terminals a-b. 10



11. Explain the working principle of a transformer with a neat, labelled diagram. Discuss the practical limitations of an ideal transformer. How does leakage flux affect its operation, and how is it minimized? 10

--End--